Camera Latency Experiment Procedure

Document Version: 1.0

Purpose

Provide a method for reading camera-based per-frame timestamps from ToughEye and TouchCam products.

Prerequisites

This experiment procedure requires Gstreamer v1.21, which can be downloaded from here.

Experiment Procedure

1. Utilize the provided test script as a starting point to record per-frame latency data using timestamps from the camera. Carefully read the README before using the script.
2. This data can be used to analyze latency distribution, dropped frames, etc
3. The camera must be configured to use the test computer as an NTP server

Experiment Details

A test program was developed in C# (using the gstreamer-sharp-netcore library), which is capable of successfully retrieving per frame timestamps generated at the camera.

The software used in this testing can be found in the “GSTTimestampTest.zip” file located in the following repository: Latency Analysis using Timestamps from Gstreamer.

With the test computer setup as the NTP server, for every frame we can measure the latency (camera frame time - system time) and we can check the frame-to-frame delta (both measured using camera timestamps and system timestamps).

Conclusion

The method outlined and utilized in the given test script can be used to successfully extract per-frame absolute timestamps utilizing the latest versions of GStreamer. Using these timestamps, it is possible to measure data such as per-frame latency.